

Module 3 Lesson 5

Linear Inequalities in Two Variables (7.4)

Objectives

In this lesson, we will learn how to

- graph a linear inequality in two variables;
- use mathematical models involving linear inequalities;
- graph a system of linear inequalities.

1. Linear Inequalities in Two Variables

If we change the equal sign in a linear equation

$$Ax + By = C$$

to one of the inequality symbols

$$>, <, \geq, \leq,$$

we obtain a linear inequality in two variables.

For example,

$$x + y < 2$$

and

$$3x - 5y \geq 15$$

are linear inequalities in two variables.

Solution of a Linear Inequality in Two Variables

A solution of an inequality in two variables is an ordered pair (x, y) that makes the inequality true when substituted into the inequality.

2. Graph of a Linear Inequality

The graph of a linear inequality in two variables is the set of all points whose coordinates satisfy the inequality.

Unlike a linear equation, whose graph is a line, the graph of a linear inequality is usually a half-plane.

Boundary Line

To graph a linear inequality, first replace the inequality symbol with an equal sign.

This gives the boundary line.

3. Solid Lines and Dashed Lines

The boundary line may be solid or dashed.

Boundary Line Rules

- Use a solid line for $\leq$ or $\geq$.
- Use a dashed line for $<$ or $>$.

A solid line means points on the boundary line are included in the solution set.

A dashed line means points on the boundary line are not included in the solution set.

4. Graphing a Linear Inequality in Two Variables

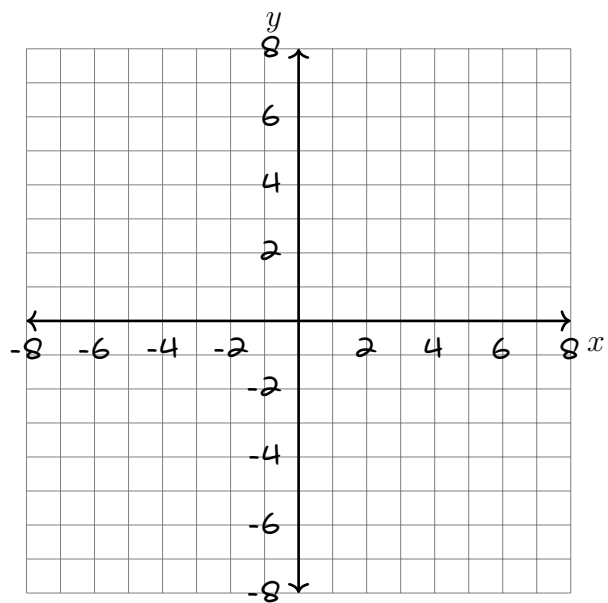
Steps for Graphing a Linear Inequality

1. Replace the inequality symbol with an equal sign and graph the boundary line.
2. Use a solid line for $\leq$ or $\geq$.
3. Use a dashed line for $<$ or $>$.
4. Choose a test point from one of the half-planes. Do not choose a point on the boundary line.
5. Substitute the test point into the original inequality.
6. If the statement is true, shade the half-plane containing the test point.
7. If the statement is false, shade the other half-plane.

5. Graphing a Linear Inequality

Example 1: Graph the inequality:

$$3x - 5y \geq 15.$$



Solution

First graph the boundary line:

$$3x - 5y = 15.$$

Find the x -intercept by setting $y = 0$:

$$3x - 5(0) = 15$$

$$3x = 15$$

$$x = 5.$$

So the line passes through

$$(5, 0).$$

Find the y -intercept by setting $x = 0$:

$$3(0) - 5y = 15$$

$$-5y = 15$$

$$y = -3.$$

So the line passes through

$$(0, -3).$$

Because the inequality is $\geq$, use a solid boundary line.

Choose the test point $(0, 0)$:

$$3x - 5y \geq 15$$

$$3(0) - 5(0) \geq 15$$

$$0 \geq 15.$$

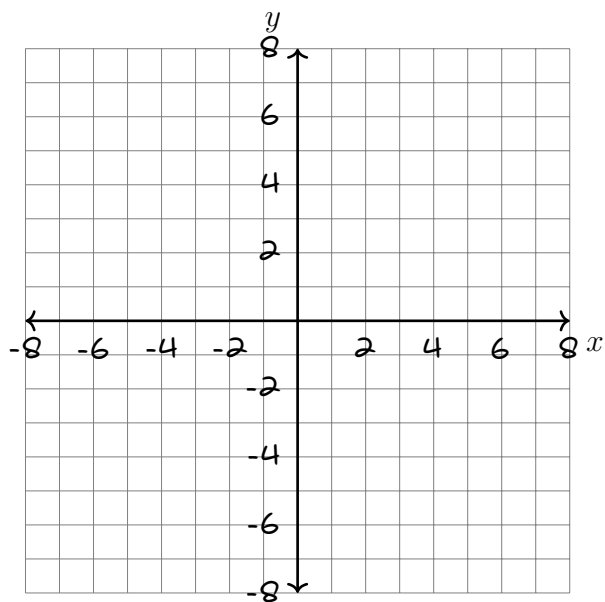
This is false.

Therefore, shade the half-plane that does not contain $(0, 0)$.

6. Another Linear Inequality

Example 2: Graph the inequality:

$$y > \frac{1}{2}x - 2.$$



Solution

First graph the boundary line:

$$y = \frac{1}{2}x - 2.$$

The y -intercept is

$$(0, -2).$$

The slope is

$$\frac{1}{2}.$$

Use the slope to find another point. From $(0, -2)$, move up 1 and right 2, giving the point

$$(2, -1).$$

Because the inequality is $>$, use a dashed boundary line.
Choose the test point $(1, 1)$:

$$1 > \frac{1}{2}(1) - 2$$

$$1 > \frac{1}{2} - 2$$

$$1 > -\frac{3}{2}.$$

This is true.

Therefore, shade the half-plane containing $(1, 1)$.

7. Graphing Without Test Points

For horizontal and vertical boundary lines, we can often shade without using a test point.

Vertical Boundary Line

For the vertical line

$$x = a :$$

- If $x > a$, shade to the right of $x = a$.
- If $x < a$, shade to the left of $x = a$.

Horizontal Boundary Line

For the horizontal line

$$y = b :$$

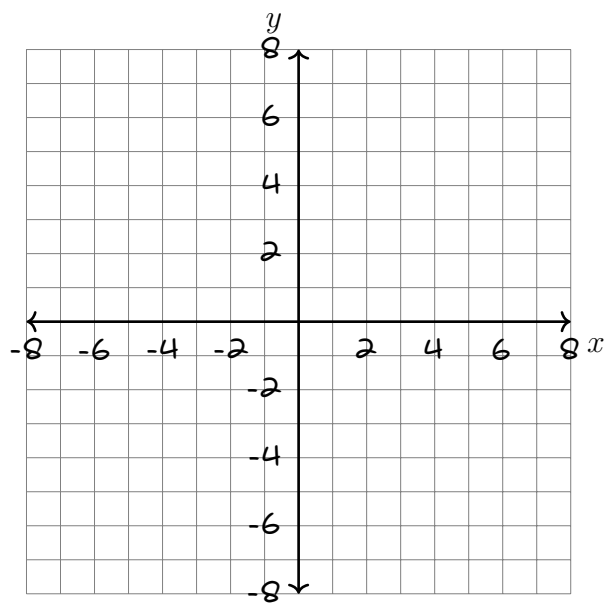
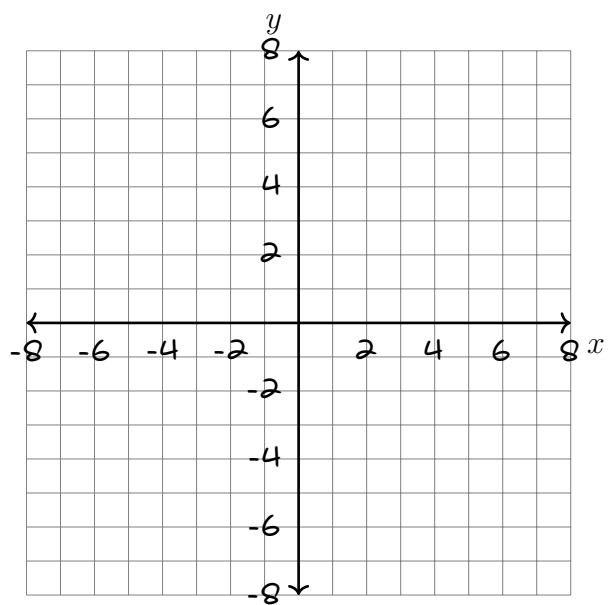
- If $y > b$, shade above $y = b$.
- If $y < b$, shade below $y = b$.

8. Horizontal and Vertical Inequalities

Example 3: Graph each inequality in a rectangular coordinate system:

$$y \leq -3$$

$$x > 2.$$



Solution

For

$$y \leq -3,$$

draw a solid horizontal line at

$$y = -3.$$

Because the inequality is $\leq$, shade below the line.

For

$$x > 2,$$

draw a dashed vertical line at

$$x = 2.$$

Because the inequality is $>$, shade to the right of the line.

9. Systems of Linear Inequalities

Just as two or more linear equations can form a system of linear equations, two or more linear inequalities can form a **system of linear inequalities**.

Solution of a System of Linear Inequalities

A solution of a system of linear inequalities in two variables is an ordered pair that satisfies each inequality in the system.

The graph of a system of linear inequalities is the region where all shaded regions overlap.

10. Graphing Systems of Linear Inequalities

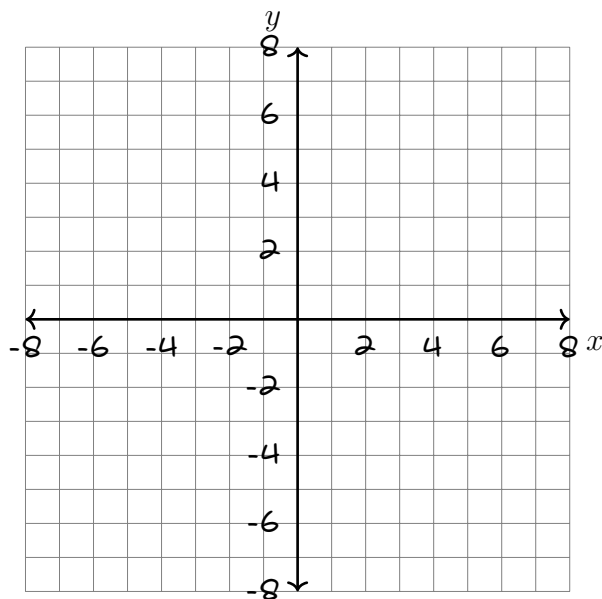
Graphing a System of Linear Inequalities

To graph a system of linear inequalities:

1. Graph each inequality on the same coordinate plane.
2. Use solid or dashed boundary lines as appropriate.
3. Shade the correct half-plane for each inequality.
4. The solution set is the overlapping shaded region.

Example 4: Graph the solution set of the system:

$$\begin{cases} x - y < 1, \\ 2x + 3y \geq 12. \end{cases}$$



Solution

First graph the boundary lines:

$$x - y = 1$$

and

$$2x + 3y = 12.$$

For

$$x - y < 1,$$

the boundary line is dashed because the inequality is $<$.

Solve for y :

$$x - y < 1$$

$$-y < 1 - x$$

Divide by -1 , reversing the inequality:

$$y > x - 1.$$

So shade above the dashed line

$$y = x - 1.$$

For

$$2x + 3y \geq 12,$$

the boundary line is solid because the inequality is $\geq$.

Solve for y :

$$2x + 3y \geq 12$$

$$3y \geq -2x + 12$$

$$y \geq -\frac{2}{3}x + 4.$$

So shade above the solid line

$$y = -\frac{2}{3}x + 4.$$

The solution set is the region where the two shaded half-planes overlap.

11. Mathematical Models with Linear Inequalities

Linear inequalities can model real-world restrictions or requirements.

For example, a situation may include limits on time, money, space, or number of items.

Common Modeling Words

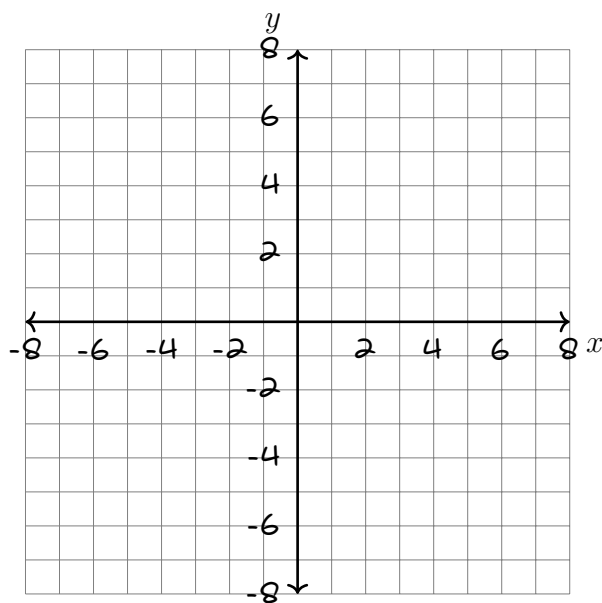
Phrase	Inequality Symbol
less than	$<$
greater than	$>$
at most / no more than	$\leq$
at least / no less than	$\geq$

Example 5: A student can spend at most \$40 on notebooks and pens.

Each notebook costs \$5, and each pen pack costs \$4.

Let x be the number of notebooks and y be the number of pen packs.

Write and graph an inequality that models this situation.



Solution

The total cost is

$$5x + 4y.$$

The student can spend at most \$40, so the inequality is

$$5x + 4y \leq 40.$$

Graph the boundary line:

$$5x + 4y = 40.$$

Find intercepts.

If $y = 0$:

$$5x = 40$$

$$x = 8.$$

So one point is

$$(8, 0).$$

If $x = 0$:

$$4y = 40$$

$$y = 10.$$

So another point is

$$(0, 10).$$

Because the inequality is $\leq$, use a solid boundary line.

Since $(0, 0)$ satisfies

$$5(0) + 4(0) \leq 40,$$

shade the half-plane containing the origin.

In context, only points with

$$x \geq 0$$

and

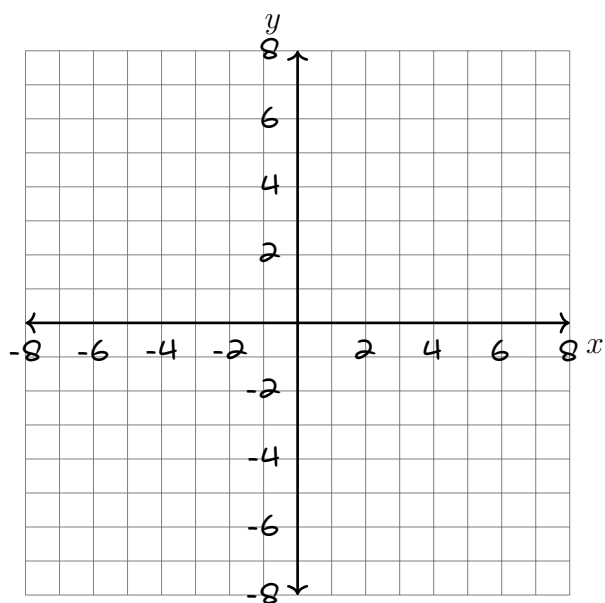
$$y \geq 0$$

make sense.

12. More Practice

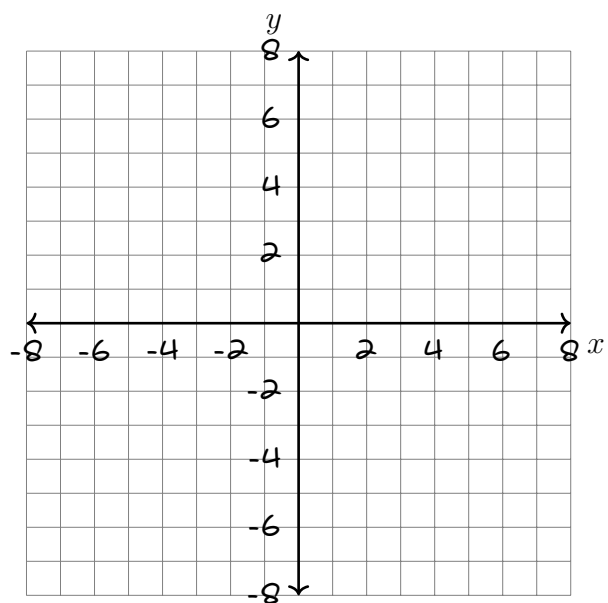
Example 6: Graph the inequality:

$$2x + y \leq 6.$$



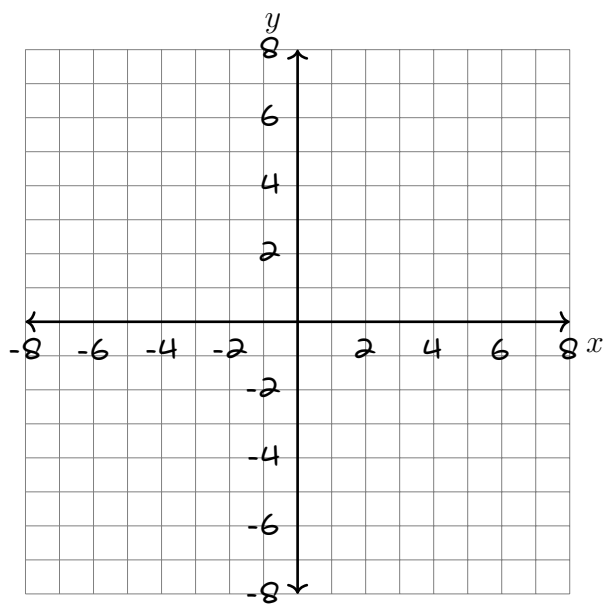
Example 7: Graph the inequality:

$$y < -2x + 3.$$



Example 8: Graph the solution set of the system:

$$\begin{cases} y \geq x + 1, \\ y < -x + 5. \end{cases}$$



Summary

Key Ideas

- A linear inequality in two variables is formed by replacing $=$ in a linear equation with $<$, $>$, $\leq$, or $\geq$.
- A solution is an ordered pair that makes the inequality true.
- The graph of a linear inequality is a half-plane.
- Use a solid boundary line for $\leq$ or $\geq$.
- Use a dashed boundary line for $<$ or $>$.
- A test point helps determine which half-plane to shade.
- A system of linear inequalities has a solution region where all shaded regions overlap.
- Linear inequalities can model real-world constraints.